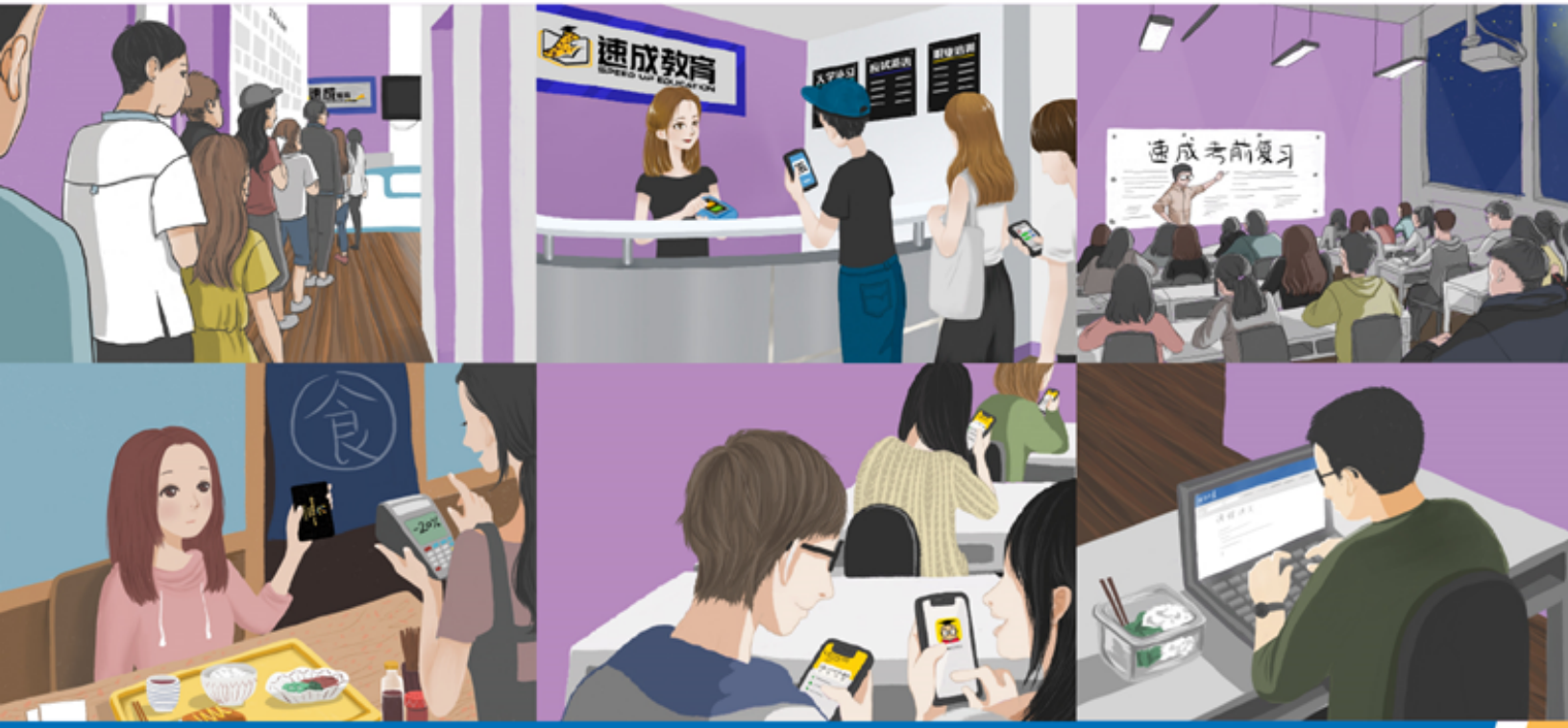
速成教育
SPEED UP EDUCATION



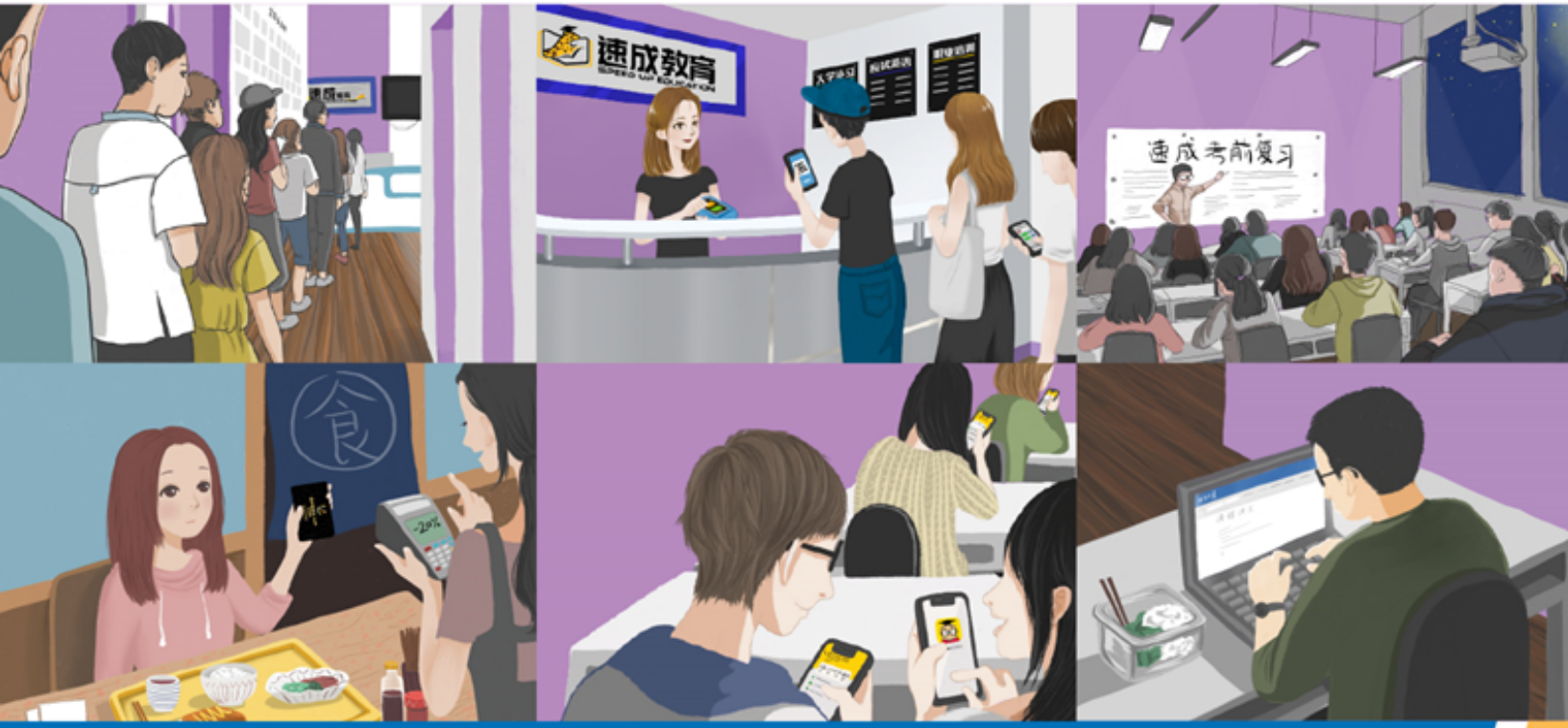
APSC 160

Introduction to Computation in Engineering Design

导师: **VC**

UBC Week 03 2026/9/26

速成教育
SPEED UP EDUCATION



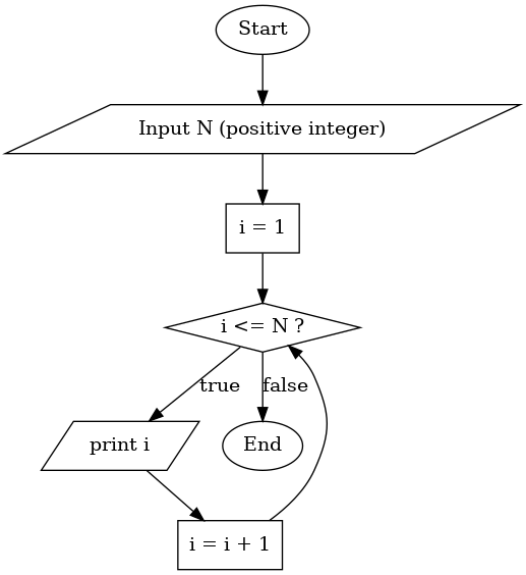
APSC 160

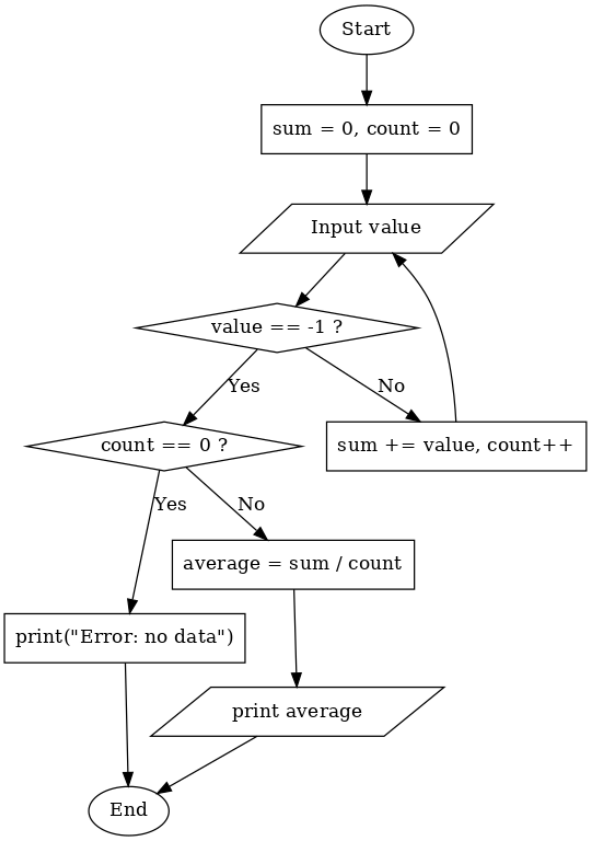
Introduction to Computation in Engineering Design

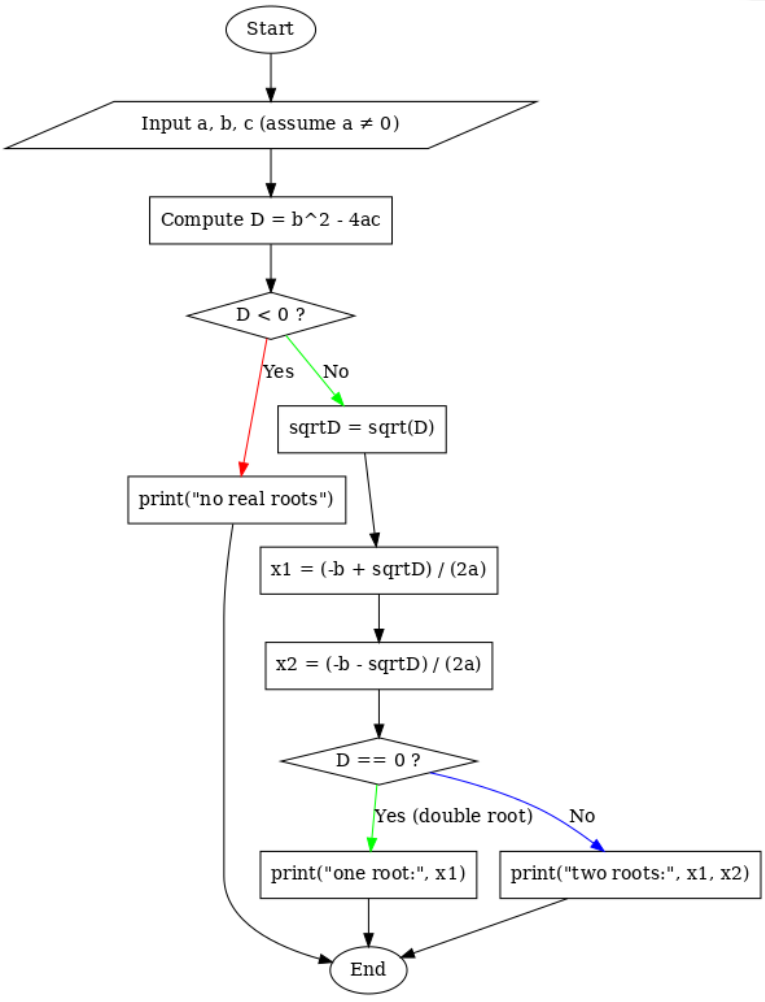
导师: **VC**

UBC Week 03 2026/9/26

Recall: Flow Chart







Loops

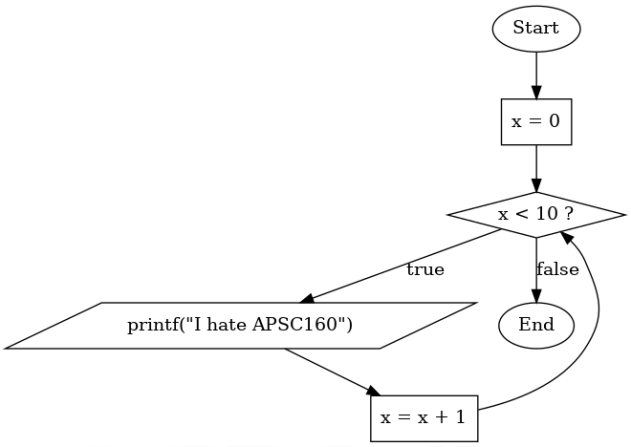
Assume we want to print 'I hate APSC160' ten times, we can repeat the code ten times:

```
printf("I hate APSC160");
printf("I hate APSC160");
printf("I hate APSC160");
...
...
printf("I hate APSC160");
```

10

A better solution will be using **while loop**

```
int x = 0;
while (x < 10) {
    printf("I hate APSC160");
    x = x + 1;
}
```



You can also change it to a **for loop**

进入循环之前 跑一次　　每次循环之前 检查

```
for(int x = 0; x < 10; x = x + 1) {
    printf("I hate APSC160");
}
```

每次循环结束 执行

```
for(int i=0; i<10; i=i+1) {
    printf("%d\n", i);
}
for(float j=10000; j>=.00001; j=j/10.0) {
    printf("%f\n", j);
}
for(char d='A'; d<'F'; d=d+1) {
    printf("%c , %d\n", d, d);
}
```

OutPut:

```
0
1
2
3
4
5
6
7
8
9
10000.000000
1000.000000
100.000000
10.000000
1.000000
0.100000
0.010000
0.001000
0.000100
A , 65
B , 66
C , 67
D , 68
E , 69
```

`x = x + 1` 可以写成 `x += 1`, 也可以写成 `x++` 或 `++x`;

那么 `x++` 和 `++x` 有何区别?

```
int a = 5;
printf("%d\n", x++);
printf("%d\n", x);
```

V.S.

```
int a = 5;
printf("%d\n", ++x);
printf("%d\n", x);
```

Short Cuts

```
var += ...
var -= ...
var *= ...
var /= ...
```

Example

```
/* Print all odd numbers from 1 to n (inclusive). */
int main() {
    int n;
    scanf("%d", &n);
```

题型: 累积 Accumulate

```
/* Print all odd numbers from 1 to n (inclusive). */
int main() {
    int n;
    scanf("%d", &n);
```

```c
/* Print the sum of all positive integers up to n (inclusive). */
int main() {
    int n;
    scanf("%d", &n);
```

```c
/* Print the sum of all integers between 1 and n that are divisible by 3 or 5.
 * Examples:
 *     if n = 10, then should print 33 (3+5+6+9+10)
 */
int main() {
    int n;
    scanf("%d", &n);

    int sum = 0;
    int i = 1;

    while (i <= n) {

       if (i % 3 == 0 || i % 5 == 0) {
          sum += i;
       }

       i++;
    }

    printf("%d\n", sum);
```

```
/* Return the number of even digits in num.
 * Examples:
 *     if num = 23641 then should print 3
 */
int main() {
    int num;
    scanf("%d", &num);

    int count = 0;
    int digit;

    while (num > 0) {
       digit = num % 10;
       if (digit % 2 == 0) {
          count += 1;
       }

       num /= 10;
    }

    printf("%d\n", count);
    }
 }
```

```
23641 % 10 => 1
23641 / 10 => 2364

2364 % 10 => 4
2364 / 10 => 236

236 % 10 => 6
236 / 10 => 23

23 % 10 => 3
23 / 10 => 2

2 % 10 => 2
2 / 10 => 0
```

```
/* Print "Paldindrome" iff the given integer <num> reads the same forwards and
 * backwards. Print "Not Paldindrome" otherwise.
 * Examples:
 *     if num = 12321 then should print "Paldindrome"
 *     if num = 1231 then should print "Not Paldinrome"
 */
int main() {
    int num;
    scanf("%d", &num);


    int digit;
    int reversed = 0;
    int original = num;

    while (num > 0) {
       digit = num % 10;
       reversed = reversed * 10 + digit;
       num /= 10;
    }


    if (reversed == original) {
       printf("Palidrome\n");
    } else {
       printf("Not Palidrome\n");
    }
```

```
23641 % 10 => 1
23641 / 10 => 2364

2364 % 10 => 4
2364 / 10 => 236

236 % 10 => 6
236 / 10 => 23

23 % 10 => 3
23 / 10 => 2

2 % 10 => 2
2 / 10 => 0
```

```
0 + 1
10 + 4
140 + 6
1460 + 3
14630 + 2
```

题型: Loop Until Stop

```
/* Print the Average of all input numbers, until the user enter -1. */
int main() {
```

更多練習

简单 ATM 模拟

写一个程式：

- 初始帐户金额 `1000`
- 使用 `while` 回圈提供选单：
  1. 存款 (Deposit)
  2. 提款 (Withdraw)
  3. 查询余额 (Balance)
  4. 离开 (Exit)
- 使用者输入选项后执行动作，直到选择 `4` 才结束。
- 提款时如果金额大于余额，输出 `Insufficient Funds`。

数字统计系统（总复习）

写一个程式：

- 使用者不断输入整数 (可以正数或负数)。
- 当输入 -9999 时停止输入。
- 输出以下统计资讯：
  1. 总和 (Sum)
  2. 平均值 (Average)
  3. 最大值 (Max)
  4. 最小值 (Min)
  5. 正数的个数与负数的个数

C 裡面有 `<limits.h>` 提供整數的範圍：
`INT_MAX` → 最大整數
`INT_MIN` → 最小整數

🎲 **挑战题：猜数字游戏 (Guess the Number Game)**

- 程式随机产生一个 1 到 100 之间的整数 (用 `rand()` 和 `srand(time(NULL))`)。
- 使用者输入数字来猜。
- 程式提示：
  - 如果猜的数字比答案 大 → 输出 "Too high!"
  - 如果猜的数字比答案 小 → 输出 "Too low!"
  - 如果猜对了 → 输出 "Correct! You got it in X tries." 然后结束。
- 使用 while 回圈来重复猜测，直到猜中为止。

题型: 提前结束

```
/* Print ‘prime’ if val is prime number, otherwise print ‘not prime’. */
int main() {
    int val;
    scanf("%d", &val);

    if (val < 2) {
       printf("Not Prime\n");
    } else {
       int i = 2;
       int is_prime = 1;

       while (i < val && is_prime) {
          if (val % i == 0) {
             is_prime = 0;
          }
          i++;
       }

       if (is_prime) {
          printf("Prime\n");
       } else {
          printf("Not Prime\n");
       }

    }
```

题型: Nested Loops

```
/* Print all prime numbers from 1 to 100. */
int main() {
```